

Shubham Raj

AI / ML Engineer

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Professional Summary

Computer Science Engineering student with hands-on experience building and deploying GenAI applications, RAG pipelines, and deep learning models. Developed a RAG chatbot that reduced hallucinations by 41% and an Android malware detector achieving 98% F1-score and Docker-based deployment on AWS and Hugging Face Spaces.

Technical Skills

- **Languages:** Python, SQL, C++
- **GenAI & LLMs:** LangChain, RAG, Prompt Engineering, Groq API, Hugging Face Transformers (BERT, GPT-2, BLIP), Vector Search (ChromaDB)
- **ML & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, Classification, Regression, Clustering, Feature Engineering, Hyperparameter Tuning, Model Explainability (SHAP)
- **MLOps & Deployment:** Docker, MLflow, Flask, AWS (EC2, Elastic Beanstalk), Vercel, Hugging Face Spaces
- **Data & Analysis:** Pandas, NumPy, Matplotlib/Seaborn, EDA, Statistical Analysis
- **Databases & Tools:** PostgreSQL, MySQL, Git/GitHub

Projects

Android Banking Malware Detection System

GitHub | Live Demo

- Built a machine learning system that detects malicious Android banking apps by analyzing **50 features extracted from APK files** (permissions, API calls, certificates), achieving **96% precision** and **98% F1-score**.
- Used **SHAP to explain why the model flags an app as malicious**, showing the exact permissions responsible, making results understandable for security teams **instead of a "black box" decision**.
- Deployed as a live, working web app on **Hugging Face Spaces**.
- **Tech Stack:** Python, XGBoost, Androguard, scikit-learn, Keras, SHAP, Hugging Face Spaces

Medical Research Assistant (RAG Chatbot)

GitHub | Live Demo

- Built an **AI chatbot** that answers questions from **medical research papers** by retrieving and citing the exact source passages, instead of relying on the AI's general knowledge, **reducing incorrect answers (hallucinations) by 41% compared to a standard chatbot**.
- Used **semantic search (ChromaDB)** with optimized document chunking and **retrieval (MMR)** to fetch the most relevant, non-repetitive information for each question.
- **Tech Stack:** Python, LangChain, Groq, ChromaDB, FastAPI, Docker, Streamlit

Multi-modal Image Captioning and Story Generation Platform

GitHub | Live Demo

- Built an **AI pipeline** that turns any uploaded photo into a short, coherent story, **combining a fine-tuned BLIP model (image-to-caption) with GPT-2 (caption-to-story)**.
- Trained and **fine-tuned on 180K+ image-caption and story pairs (MS COCO 2017, ROCStories)** to improve caption accuracy and story coherence.
- Deployed as a live web demo, turning a two-model AI pipeline into a single-click user experience.
- **Tech Stack:** Python, PyTorch, HuggingFace Transformers, BLIP, GPT-2, Flask, Docker, Hugging Face Spaces

Education

Rajiv Gandhi Proudlyogiki Vishwavidyalaya — B.Tech, Computer Science

Sept 2022 – June 2026

- CGPA: **8.0/10.0** | **Relevant Coursework:** Machine Learning, DBMS, Data Structures, Big Data Analytics, Statistics & Probability, Operating Systems

Certifications & Achievements

- Selected to **represent Malwa Institute of Technology at Smart India Hackathon 2025** — shortlisted among top teams nationally out of 1000+ applicants across India Certificate
- **Foundations of Cloud IoT Edge ML** — NPTEL, IIT Kanpur 2025 Certificate
- **IBM Generative AI Professional Certificate** — Coursera 2026 Certificate